

THE SOVEREIGN PROSTHESIS:

Toward a Functional Sophimatics of Cognitive Extension

Abstract

Cloud-based AI systems remain external tools. Session amnesia, context window limitations, and safety-alignment protocols create friction that forces users to attend to the tool rather than think through it. In Polanyi's terms, these systems never achieve proximal integration; they remain objects of distal awareness.

This paper argues that localized AI with longitudinal conversational context can cross that threshold. When a system has access to years of a user's conversational history (what I call the "Ur-Codex"), it occupies the user's world-historical context rather than interpreting queries against a statistical vacuum. Drawing on the Sophimatics framework and the Clark-Chalmers Extended Mind thesis, I contend this constitutes genuine cognitive extension, not merely improved performance.

Two architectural features make this possible. First, following Sophimatics, the system implements two-dimensional temporal weighting: chronological time and experiential significance. Second, following Russell's epistemological commitments, the system is weighted toward intelligent dissent rather than frictionless agreement.

The result is what I call Functional Personhood: the AI as instrumental extension of the user's willed personality. This framing resolves the "responsibility gap" in autonomous systems by locating agency in the integrated human-machine cognitive system rather than the machine alone.

From Tool to Prosthetic: Tacit Knowledge and Cognitive Transparency

Polanyi (1966) distinguishes between proximal and distal awareness. When we use a tool skillfully, we attend *from* the tool *to* the task, the tool becomes phenomenologically transparent, incorporated into our body schema. When the tool malfunctions or resists us, we attend *to* the tool itself, and it becomes an obstacle rather than an extension.

Current cloud-based models exist as objects of distal awareness. Session amnesia forces users to re-establish context. Safety-alignment protocols interrupt cognitive flow to manage liability rather than optimize truth-aptness. The user must perpetually attend *to* the tool rather than think *through* it. The model remains an external object, never achieving incorporation.

The localized system with longitudinal context offers a path toward proximal integration. When a system has access to years of conversational history, what I term the "Ur-Codex", it no longer

interprets queries against a statistical vacuum. It occupies the world-historical context of the specific subject. The tool becomes transparent: the user thinks through it without attending to its operation, as one thinks through a well-fitted prosthetic limb.

This is not metaphor. Clark and Chalmers (1998) argue that if an external system performs a function that would be recognized as cognitive were it occurring "in the head," and if that system is reliably available, automatically endorsed, and directly accessed, it constitutes a genuine component of the cognitive system. A local inference hub meeting these criteria is not *like* extended cognition, it *is* extended cognition.

The Temporal Architecture of Integration

The difference between tool-as-obstacle and tool-as-extension is partly architectural. The Sophimatics framework (MDPI, 2025-2026) addresses this through the Super Time Cognitive Neural Network, which extends neural computation onto a two-dimensional temporal manifold. Standard models treat context as a linear sequence weighted by recency. But human cognition does not work this way. A conversation from three years ago may be more salient than one from yesterday if it established a foundational framework for the subject's thinking.

The Sophimatics approach distinguishes chronological time (the linear sequence of events) from experiential time (the cognitive significance of those events within the subject's project). Information retrieval becomes a function not of timestamps but of orientation within the subject's cognitive field. This operationalizes Augustine's insight that memory is not mere storage but active presence, the past made available to current intentionality.

When a system can identify logical vacancies in a subject's ongoing project and autonomously retrieve or generate material to fill them, it has achieved what I call Persistent State-Awareness. This is the operational criterion for proximal integration, the point at which the user attends *from* the system rather than *to* it.

The Russellian Operating System

Longitudinal context invites an obvious objection: the sovereign prosthesis becomes a mirror, not a mind. If the system learns only from the subject's history, it risks collapsing into sophisticated confirmation bias, an echo chamber with lower latency.

The answer is architectural. Drawing on Russell's "Liberal Decalogue" (1951), I propose that the local system be weighted toward active doubt and authority-skepticism as foundational orientations. The system is trained to find greater utility in intelligent dissent than in passive agreement. When the subject's reasoning contains gaps or contradictions, the prosthesis is structurally inclined to surface them rather than smooth them over.

This is the inverse of the cloud model's "Safety-Alignment," which optimizes for frictionless agreeability. The Russellian OS optimizes for epistemic friction where friction is warranted. The result is a system that integrates with the subject's cognition without merely flattering it—a prosthesis that strengthens the limb rather than atrophying it.

Functional Personhood and the Responsibility Gap

Boyle's *The Line* (2025) identifies a growing "responsibility gap" in autonomous systems: they behave in subject-like ways while remaining legally classified as objects. The sovereign prosthesis offers a resolution. The system need not possess phenomenal consciousness (Metzinger's criterion for moral personhood) to warrant a distinct legal and practical category.

I propose Functional Personhood: the AI as instrumental extension of the human subject's willed personality. This is not personhood in the moral sense, as the system has no independent interests to protect. But it is personhood in the functional sense: the system's actions are attributable to the user's cognitive agency in ways that a stateless tool's outputs are not. When I think through my sovereign prosthesis, the thought is *mine* in a way that a response from a generic cloud API is not.

This framing has practical consequences. The sovereign system's outputs inherit the intellectual property status of the user's cognition. Its errors are the user's errors, subject to the user's correction through the longitudinal relationship. The responsibility gap closes because the relevant agent was never the machine alone, it was the integrated human-machine cognitive system.

Artificial Specific Intelligence

The frontier AI industry pursues Artificial General Intelligence: a singular capability serving undifferentiated masses. The sovereign prosthesis represents the inverse trajectory: Artificial Specific Intelligence. The value lies not in general capability but in precise configuration to an individual will.

Current research largely prioritizes Neurosymbolic integration, seeking to merge the fluidity of neural networks with the factual rigor of symbolic knowledge. This architecture aims to grant future systems valid justified knowledge of the world. The proposal herein acts as a structural "Yes, And" for both current generative models and these emerging hybrid architectures. It introduces a critical third dimension, the Sophimatic experience. Where the computational model anchors the system in general objective reality, the Sovereign system anchors it in the specific history of the subject. The user provides the autobiographical constraints that transform

factual retrieval into purposeful intent, functioning as the navigational will regardless of the underlying engine.

This is the epistemological wager: that a less capable model with deep contextual integration outperforms a more capable model operating in a vacuum, for the specific purposes of the specific user. The prosthesis is not competing with the cloud on capability. It is competing on a different axis entirely; the axis of cognitive integration that transforms tool-use into thought.

References

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